Extended Gauge-Fixed Standard Model Lagrangian

Methodology, Field Bundles, and Sign Conventions

1 Introduction and Model Scope

This reference sheet compiles the mathematical foundations, geometric definitions, and conventions for the Extended Standard Model (ESM). The specification extends the minimal Standard Model field content by including right-handed sterile neutrinos ν_R^λ with Dirac masses m_ν^λ and no Majorana mass terms.

All physical fields are smooth sections, denoted by Γ, of appropriate vector bundles over flat Minkowski spacetime $\mathcal{M} \cong \mathbb{R}^{1,3}$. All bundles are trivial over $\mathcal{M}$; in the chosen coordinates and bundle trivializations, ∂_μ acts componentwise on field components.

2 Vector Bundles and Field Classifications

Every physical and auxiliary field is classified as a section of a specified bundle over $\mathcal{M}$.

Table 1: Geometric classification of physical and auxiliary fields

Field category	Symbol(s)	Mass dim.	Vector bundle or space
Scalars (real)	H, ϕ^0	1	$\mathcal{M} \times \mathbb{R}$ (trivial real line bundle)
Scalars (complex)	$\phi^\pm$, $(\phi^+)^* = \phi^-$	1	$\mathcal{M} \times \mathbb{C}$ (trivial complex line bundle)
Gauge fields ($SU(3)_c$)	g_μ^a	1	$T^*\mathcal{M}$, with $ig_\mu \in \Gamma(T^*\mathcal{M} \otimes \mathfrak{su}(3))$
Gauge fields ($SU(2)_L$)	W_μ^a, $a = 1, 2, 3$	1	Real cotangent bundle $T^*\mathcal{M}$
Gauge field ($U(1)_Y$)	B_μ	1	Real cotangent bundle $T^*\mathcal{M}$
Physical neutral gauge fields	Z_μ, A_μ	1	Real cotangent bundle $T^*\mathcal{M}$
Physical charged gauge fields	W_μ^+, $W_\mu^- = (W_\mu^+)^*$	1	Complex cotangent bundle $T^*\mathcal{M} \otimes \mathbb{C}$
Fermion fields (Dirac)	$e^\lambda, \nu^\lambda, u_j^\lambda, d_j^\lambda$	3/2	Spinor bundle S over $\mathcal{M}$, acting on $\mathbb{C}^4$
Colour ghosts	$G^a, \bar{G}^a$	1	$\mathcal{M} \times \Lambda_{\text{odd}}$ with $SU(3)_c$ adjoint structure
Electroweak ghosts	$X^\pm, \bar{X}^\pm, X^0, \bar{X}^0, Y, \bar{Y}$	1	$\mathcal{M} \times \Lambda_{\text{odd}}$

2.1 Sterile Neutrino Details

The neutrino fields are Dirac:

$$\nu^\lambda = \nu_L^\lambda + \nu_R^\lambda, \qquad \nu_{L,R}^\lambda = P_{L,R}\nu^\lambda. \tag{1}$$

The right-handed component ν_R^λ is a singlet under $SU(3)_c \times SU(2)_L \times U(1)_Y$ and therefore has no gauge-boson coupling. It enters through its kinetic term, the Dirac mass m_ν^λ, and the H, ϕ^0 and $\phi^\pm$ Yukawa couplings.

3 Sign and Parameter Conventions

The following definitions fix the sign and parameter conventions used throughout the gauge-fixed specification.

Table 2: Sign conventions and parameter definitions

Convention	Parameter or definition	Meaning
Minkowski metric	$\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1)$	Flat metric on $\mathcal{M}$; $\partial^2 \equiv \eta^{\mu\nu}\partial_\mu\partial_\nu$.
Colour covariant derivative	$\sigma_s = -1$	$D^{(c)}_\mu q = (\partial_\mu - ig_s T^a g^a_\mu)q$.
Higgs covariant derivative	$Y_\Phi = +1$	$D_\mu\Phi = (\partial_\mu - igT^a_L W^a_\mu - i\frac{g'}{2}Y_\Phi B_\mu)\Phi$.
Electric charge	$Q = T^3_L + Y/2$	Gell–Mann–Nishijima relation.
Tadpole parameter	$\beta_h = 0$	Tree-level condition giving no tadpole term.
Gauge-fixing parameter	$\xi = 1$	’t Hooft–Feynman gauge.
Faddeev–Popov sign	$\sigma_G = +1$	$\mathcal{L}_{\mathrm{FP}} = \sigma_G \bar{c}_I(\delta F_I/\delta\theta_J)c_J$.

The weak mixing parameters obey $s_w^2 + c_w^2 = 1$, with $e = gs_w > 0$, and the Higgs vacuum convention is $v = 2M/g$.

4 Index Domains and Summation Rules

- **Colour ($SU(3)_c$):** adjoint indices $a, b, c, r, s \in \{1, \ldots, 8\}$; fundamental indices $i, j \in \{1, 2, 3\}$.
- **Weak isospin ($SU(2)_L$):** fundamental indices $\alpha, \beta \in \{1, 2\}$.
- **Weak-isospin adjoint:** $a, b, c \in \{1, 2, 3\}$ when attached to W^a_μ, θ^a, T^a_L or ϵ^{abc}.
- **Generation (flavour):** $\lambda, \kappa \in \{1, 2, 3\}$.

4.1 Summation Notation Control

Repeated Lorentz, colour, adjoint, weak-isospin and generation indices are summed over their declared domains unless explicitly stated otherwise. Generation labels carried by diagonal mass parameters $(m^\kappa_e, m^\lambda_\nu, m^\lambda_u, m^\lambda_d)$ select the mass associated with the accompanying field index; they do not introduce an additional contraction. Such a label may therefore appear more than twice in one monomial without implying a further sum.

5 Complete Infinitesimal Gauge Transformations

All Faddeev–Popov ghost vertices are obtained by varying the gauge-fixing functions under the complete infinitesimal transformations below.

Gauge-basis transformations

$$\delta g^a_\mu = \partial_\mu\theta^a_g + g_s f^{abc} g^b_\mu \theta^c_g, \tag{2}$$

$$\delta W^c_\mu = \partial_\mu\theta^c + g\epsilon^{abc} W^a_\mu \theta^b, \qquad c = 1, 2, 3, \tag{3}$$

$$\delta B_\mu = \partial_\mu\theta_Y. \tag{4}$$

Physical gauge-boson basis

$$\delta W^\pm_\mu = \partial_\mu\theta^\pm \pm ig\Big[(c_w\theta_Z + s_w\theta_A)W^\pm_\mu - (c_w Z_\mu + s_w A_\mu)\theta^\pm\Big], \tag{5}$$

$$\delta Z_\mu = \partial_\mu\theta_Z + igc_w(W^-_\mu\theta^+ - W^+_\mu\theta^-), \tag{6}$$

$$\delta A_\mu = \partial_\mu\theta_A + igs_w(W^-_\mu\theta^+ - W^+_\mu\theta^-). \tag{7}$$

5.1 Scalar Transformations

Using $\Phi = (\phi^+, (v + H + i\phi^0)/\sqrt{2})^T$ and $v = 2M/g$, the scalar transformations are

$$\delta\phi^\pm = \pm iM\theta^\pm \pm \frac{ig}{2}\theta^\pm(H \pm i\phi^0) \pm i\left[\frac{g(2c_w^2 - 1)}{2c_w}\theta_Z + gs_w\theta_A\right]\phi^\pm, \tag{8}$$

$$\delta\phi^0 = -\frac{M}{c_w}\theta_Z - \frac{g}{2c_w}\theta_Z H + \frac{g}{2}(\theta^-\phi^+ + \theta^+\phi^-), \tag{9}$$

$$\delta H = \frac{g}{2c_w}\theta_Z\phi^0 + \frac{ig}{2}(\theta^-\phi^+ - \theta^+\phi^-). \tag{10}$$

These relations follow from

$$\delta\Phi = i\left(gT_L^a\theta^a + \frac{g'}{2}\theta_Y\right)\Phi.$$

Together with the gauge-fixing functions, they generate the ghost–scalar interactions. In particular, the θ_Z and θ_A terms in $\delta\phi^\pm$ generate the $\bar{X}^\pm X^0\phi^\pm$ and $\bar{X}^\pm Y\phi^\pm$ vertices, while the $\theta^\mp\phi^\pm$ terms in $\delta\phi^0$ generate the $\bar{X}^0 X^\mp\phi^\pm$ vertices.

6 Gauge-Fixing Functions

For completeness, the $\xi = 1$ gauge-fixing functions used with the transformations above are

$$F_+ = \partial_\mu W^{+\mu} - iM\phi^+, \qquad F_- = \partial_\mu W^{-\mu} + iM\phi^-,$$

$$F_Z = \partial_\mu Z^\mu - \frac{M}{c_w}\phi^0, \qquad F_A = \partial_\mu A^\mu, \qquad F_g^a = \partial_\mu g^{a\mu}.$$

The Faddeev–Popov operator is $\mathcal{M}_{IJ} = \delta F_I/\delta\theta_J$, evaluated at zero gauge parameters.

This sheet summarizes the definitions used by `SM_RHNu_Dirac_v1.tex`. In any conflict, that complete specification is the authoritative source.